

Pilot Assignment in Cell-Free Massive MIMO

System Model, Pilot Contamination, and Adaptive Assignment

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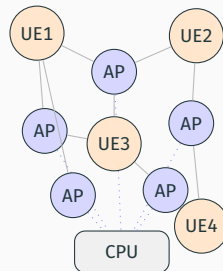
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Basic idea

- Many distributed APs connect to a central processing unit.
- From the UE side, there is no fixed cell boundary.
- A UE can be served by several APs.
- An AP can receive signals from several UEs.

Implication

- Macro-diversity improves coverage.
- Many AP-UE channel links must be estimated.
- Pilot assignment becomes a network-level problem.



Channel between AP l and UE k

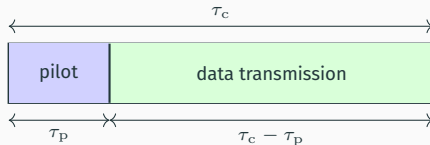
$$\mathbf{h}_{lk} = \sqrt{\beta_{lk}} \mathbf{g}_{lk}, \quad \mathbf{g}_{lk} \sim \mathcal{CN}(\mathbf{0}, \mathbf{I}_N)$$

- L APs, K UEs, N antennas per AP.
- β_{lk} captures pathloss and shadowing.
- Dominant APs/beams are the useful structure for pilot assignment.

Spectral efficiency form

$$\text{SE}_k = \left(1 - \frac{\tau_p}{\tau_c}\right) \log_2(1 + \text{SINR}_k)$$

Coherence block



Key tension

More pilot symbols can reduce reuse pressure, but they leave fewer symbols for data.

Pilot observation at AP l

$$\begin{aligned}\mathbf{Y}_l^p &= \sum_{i=1}^K \sqrt{p_i} \mathbf{h}_{li} \phi_{t_i}^T + \mathbf{N}_l, \\ y_{lk}^p &= \mathbf{Y}_l^p \phi_{t_k}^* \\ &= \sqrt{p_k} \tau_p \mathbf{h}_{lk} + \sum_{i \in \mathcal{P}_k \setminus \{k\}} \sqrt{p_i} \tau_p \mathbf{h}_{li} + \mathbf{n}_{lk}. \\ \mathcal{P}_k &= \{i : t_i = t_k\}\end{aligned}$$

Interpretation

- UE k transmits pilot ϕ_{t_k} .
- AP l correlates the received signal with that pilot.
- If another UE uses the same pilot, its channel enters the same statistic.
- The estimate is then contaminated before data detection starts.

Pilot contamination

Same pilot plus overlapping strong channels creates mixed channel estimates.

Why assignment is difficult

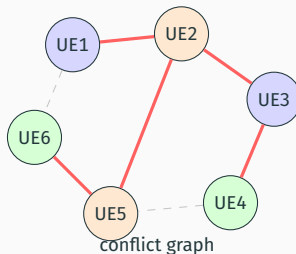
- Orthogonal pilot count is limited: usually $\tau_p < K$.
- Pilot reuse is therefore necessary.
- Each UE can overlap with many other UEs across APs and beams.
- A high-overlap same-pilot pair lowers channel-estimation quality and SINR.

Graph view

- Node = UE.
- Edge = strong effective conflict.
- Color = pilot index.

Research question

Which UEs can reuse a pilot without creating a strong effective conflict?



fewer colors mean fewer actual pilots

$$SE_k = \left(1 - \frac{\tau_p}{\tau_c}\right) \log_2(1 + \text{SINR}_k)$$

More pilots

- More orthogonal pilot resources.
- Lower pilot reuse pressure.
- Higher training overhead.
- Less data time in the coherence block.

Fewer pilots

- Larger prelog factor.
- Better data-time efficiency.
- Higher contamination risk.
- Works only if the retained structure keeps SINR strong.

Main lens

A useful assignment should reduce pilot overhead without throwing away too much channel quality.

Gao et al.

- AP-domain pilot assignment.
- Many-to-many matching idea.
- Strong small-pilot regime result.
- Pilot count is a fixed design input.

Mussbah et al.

- Beam-domain active/moderate beam graph.
- Conflict graph colored by DSATUR.
- Actual pilot count can become adaptive.
- RF-chain activity matters for EE.

Family	Domain	Conflict signal	Pilot-count rule
Gao	AP / large-scale fading	UE-AP matching	fixed design budget
Mussbah	beam domain	active/moderate beam overlap	graph chromatic number
Proposed set	AP + beam variants	sparsified effective conflicts	adaptive where possible

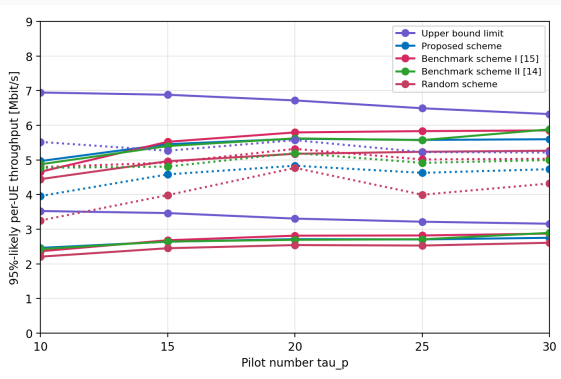
What this slide supports

- 200 Monte Carlo setups.
- $\tau_p = 10$, max-min power.
- Gao vs Random: **+21.7%**.
- Paper claim: approximately +23%.

How to use it

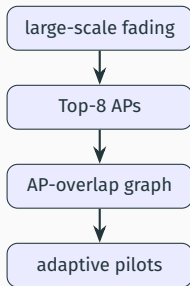
- Use as a reproduction anchor.
- Do not use it as the final multi-antenna benchmark.

Environment: Gao reproduction setting; different from the common-ground multi-antenna results.



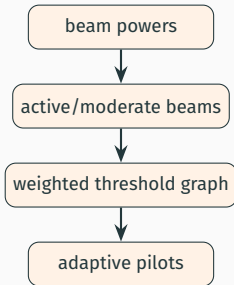
AP-Top8 Adaptive

- Strongest 8 APs per UE.
- Conflict by shared Top-8 APs.
- Adaptive graph coloring.



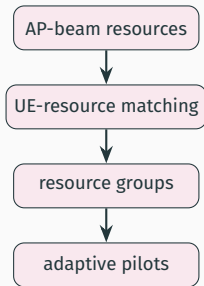
Beam-Weighted Threshold

- Active/moderate beam split.
- Higher weight for active-active overlap.
- Threshold weak conflicts.



Beam-Resource Matching

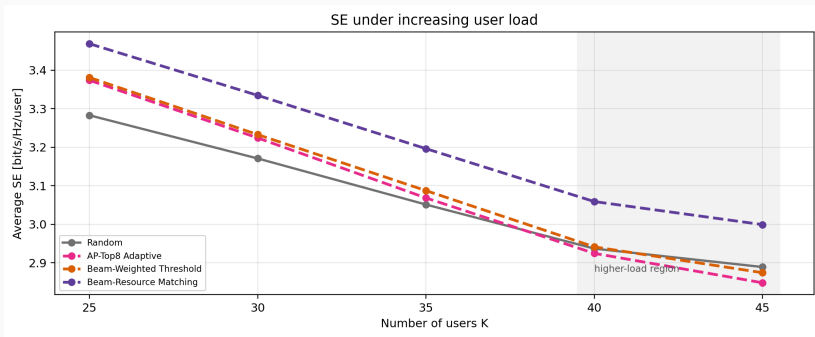
- Candidate AP-beam resources.
- Many-to-many UE-resource matching.
- Adaptive pilot grouping.



Method	Domain	Conflict signal	Pilot count	Role
Random	AP	none	fixed	reference
Gao Matching	AP	UE-AP matching	fixed	reference
Mussbah Beam Graph	beam	binary beam overlap	adaptive	reference
AP-Top8 Adaptive	AP	Top-8 AP overlap	adaptive	proposed
Beam-Weighted Threshold	beam	weighted beam overlap	adaptive	proposed
Beam-Resource Matching	beam	AP-beam resource matching	adaptive	proposed

Unifying view

Different conflict signals, same lever: reduce effective conflicts while keeping dominant channel structure.



Environment: $K = 25, \dots, 45$, $L = 100$, $N = 8$,
 $\tau_c = 100$, $\tau_p^{design} = 10$, 200 setups, closed-form SE.

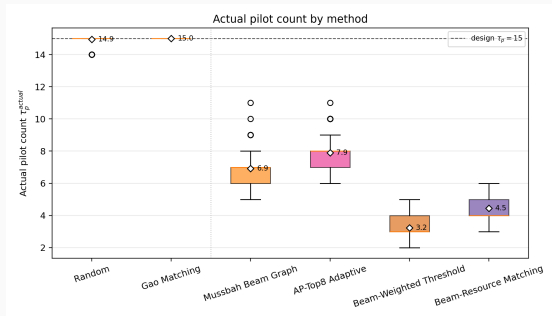
Reading: AP-Top8 and Beam-Weighted cross below Random at high load; Beam-Resource Matching remains positive at $K = 45$ (+3.81%).

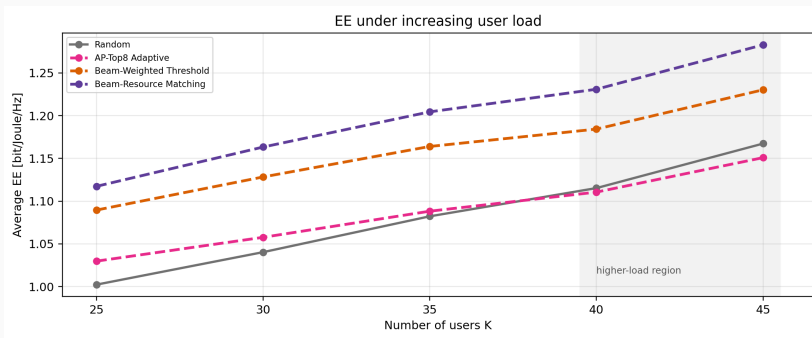
Common-ground snapshot

Method	$\bar{\tau}_p$	prelog ratio	SE gain
Random	14.94	baseline	baseline
Beam-Weighted Threshold	3.23	+8.67%	+8.53%
Beam-Resource Matching	4.46	+7.76%	+7.80%
AP-Top8 Adaptive	7.90	+5.21%	+5.28%

Reading

- Fewer pilots improve the prelog factor.
- Too much pruning can damage SINR under high load.
- The useful target is balance, not the minimum possible pilot count.





Environment: $L = 100$, $N = 8$, $\tau_c = 100$, $\tau_p^{design} = 10$, 200 setups, closed-form SE/EE proxy.

At $K = 45$ vs Random: Beam-Resource **+9.90%**, Beam-Weighted **+5.39%**, AP-Top8 **-1.41%**.

Pilot assignment is a balance problem.

Prune enough to reduce pilots, but not enough to destroy SINR.

What we can defend

- Gao gives a useful reproduction anchor.
- Multiple proposed methods sparsify conflicts.
- User-load crossover shows that pruning can overreach.
- Beam-Resource Matching is the most robust current direction.

What we should not oversell

- Fewer pilots are not always better.
- $\tau_p = 1$ is not the design conclusion.
- Cross-environment absolute SE should not be mixed.
- EE is a simplified proxy.

Final claim: good UE clustering and AP/beam pruning should balance SINR preservation against pilot-count reduction.

Backup: Closed-Form Environment And Load Sensitivity

Separate closed-form environment

- $K = 30, L = 100, N = 8$.
- $\tau_c = 100, \tau_p^{design} = 10$.
- 200 setups.
- Closed-form MMSE+MRC SE.

Method	mean SE	$\bar{\tau}_p$
Random	3.171	10.00
AP-Top8 Adaptive	3.224	8.34
Beam-Weighted Threshold	3.233	7.42
Beam-Resource Matching	3.335	4.59

Closed-form K-sweep

K	AP-Top8	Weighted	Resource matching
25	+2.77%	+2.99%	+5.66%
30	+1.68%	+1.96%	+5.18%
35	+0.57%	+1.19%	+4.76%
40	-0.42%	+0.15%	+4.16%
45	-1.42%	-0.51%	+3.81%

- AP-domain Top-N weakens as load grows.
- Beam-Resource Matching remains strongest in this sweep.

Backup: Graph Consistency Rules

- 1 Use common-ground MC figures for the main ranking.
- 2 Use method names, not person-specific labels.
- 3 Use only Random, Gao Matching, and Mussbah Beam Graph as main references.
- 4 Use Gao reproduction only as a small-pilot reproduction anchor.
- 5 Say “EE proxy” for the current EE plots.
- 6 Do not claim a $\tau_p = 7$ optimum from the current tau sweep.
- 7 Add bootstrap CI for the final 12-scheme run before using significance language.